The Ephemeris

September 2019
Volume 30 Number 03- The Official Publication of the San Jose Astronomical Association

September 2019 - December 2019 Events

Board & General Meetings
Saturday 9/14, 10/12, 11/9, 12/14
Board Meetings: 6 -7:30pm
General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)
Sunday 9/8, 10/6, 11/3, 12/1

Solar Sunday (1:30-3:30pm)
Sunday 9/8, 10/6, 11/3, 12/1

Intro to the Night Sky Class &
Houge Park 1Q In-Town Star Party
Friday 9/6, 10/4, 11/1, 12/6

Houge Park 3Q In-Town Star Party
Friday 9/20, 10/18, 11/15, 12/20

RCDO Starry Nights Star Party
Saturday 9/21, 10/19, 11/23, 12/21

Imaging SIG Mtg
Tuesday 9/17, 10/15, 11/19, 12/17

Astro Imaging Clinics
(at Little Uvas OSP)
Saturday 9/28, 12/28

Astro Imaging Workshops
(at Little Uvas OSP)
Saturday 9/28, 10/26, 11/23

Quick STARt
By appointment

Binocular Star Gazing
Saturday (will resume in Spring 2020)

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

INSIDE THIS ISSUE

Photo by SJAA Member: Alan Pham
Jan 21, 2019
Location: San Jose, CA

Introducing Cat Cvengros Second Harvest Food Bank

Credits: Vikas Kache, Cat Cvengros, and Silicon Valley Business Journal

Cat Cvengros receives the award on July 26, 2019

Pictured above l to r: Mary Huss—President of Silicon Valley Business Journal; Cat Cvengros and J Jennings Moss—editor-in-chief of Silicon Valley Business Journal

Cat Cvengros was the recipient of the Silicon Valley Business Journal's 40 Under 40 Silicon Valley Award, an honor for young professionals in Silicon Valley who have distinguished themselves through their work and community involvement. Cat is the Vice President of Second Harvest of Silicon Valley, where she has grown revenues for the food bank and expanded its impact. She's grown revenues for children's programming by 27%, and revenues overall by 9%. She's an accomplished public speaker and storyteller and is sought after nationwide to present on topics related to ending poverty and the power of stories to change lives. In addition to her professional accolades, Cat was honored for her work with SJAA, where she coordinates the School Star Party Program. We asked her a few questions about the program here:

Can you give a background on how you got started with the school star party program?

I read about it in the Ephemeris not too long after I joined the club. I thought helping out with such an established program would be fun and be an easy way for me to use my skills to help kids learn about the universe.

How many schools are participating? Any idea about how many students have been reached out to?

It varies from year to year, but this year we have requests from about 20 schools, and 12 school star parties are already scheduled. Last year we had 39 volunteer astronomers educate about 885 kids throughout our community. It was a slower year as we had a smoky fall due to the fire in Paradise and then a rainy spring, both of which limited our events.

What level of engagement are you seeing from the students? What are some common questions they have?

The best moment is showing a child the craters on the moon or the rings of Saturn for the first time. They exclaim with wonder and come back over and over. Each school has different science programs, and that is reflected in the questions we get at star parties. Some students understand light years to some degree and ask about what gases certain stars emit. Other students may be seeing a telescope for the first time and are unsure of how many planets we have. The star parties give us an opportunity to make science and astronomy approachable for everyone, and the parents always share in the delight with their kids.

What are some of your future plans for the school star party program?

I hope to serve more kids this year; we already have more school star parties scheduled than we had last year. Jim Van Nuland ran this program with such passion and care for many years, and it is an honor to get to carry on the torch.

From the Silicon Valley Business Journal July 26th 2019 issue

When someone lists "public transit" as their dream car — as Cat Cvengros did in a recent questionnaire — serving the greater good is obviously a way of life. From her two-year stint in the Peace Corps to her current role as vice president of development and marketing at Second Harvest Food Bank, the Michigan native has walked the walk.

At Second Harvest, that means boosting revenue by 27 percent for programs that reduce childhood hunger since joining the organization in early 2017 while overseeing a volunteer workforce that collectively gives more than 300,000 hours a year.

During her previous job as chief development officer at Santa Rosa's Social Advocates for Youth, she doubled the group's operating revenue over a five-year stretch while heading a $9.8 million fundraising campaign to open the Dream Center youth campus.

On the side, Cvengros broadens her empathetic approach, whether it be getting food to Mineta San Jose International Airport's TSA workers during the government shutdown earlier this year or acting as lead coordinator for the San Jose Astronomical Association School Star Party program.

M 35 Paper

Credit: Hitesh Dholakia

The story is personal and although somewhat OT sharing this with SJAA family.

The following paper was accepted in the Publications of the Astronomical Society of the Pacific (PASP) journal. It is about the discovery of an exoplanet in the M35 cluster region (Mind you not in M35 - more on that later).

It started as a high school project in their 10th grade. My sons (Shashank & Shishir) were curious while photographing a star-forming region. If all stars are born in clusters, would there be detectable exoplanets in open clusters? They did not realize at the time that it was an open area of research.

When in 2013, Nasa's Kepler telescope's reaction wheel failed, it had to abandon its original mission of observing the Cygnus/Lyra region. The K2 mission (somewhat hobbled Kepler) needed to observe various parts of ecliptic due to the inherent nature of its recovery. And since it was a trial mission, they opened the data to the public, and the first observations included the region of M35. They used this opportunity to write an algorithm/pipeline to search for exoplanets.

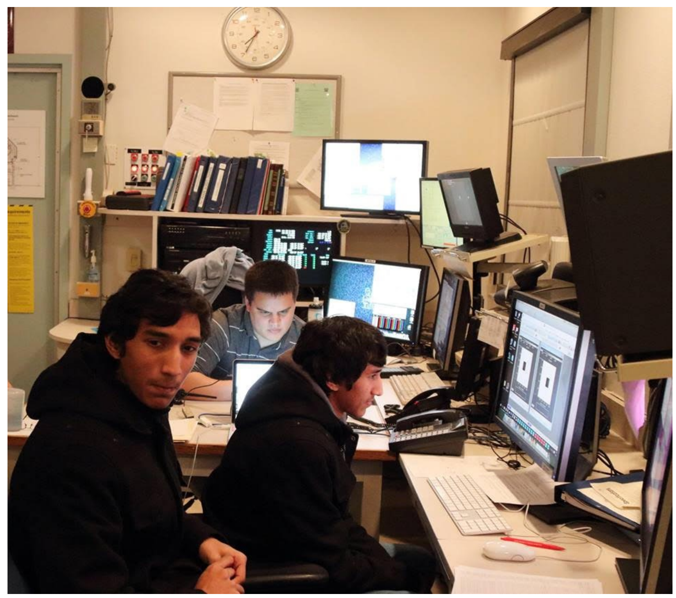
They did find an exoplanet candidate in a hot, A-type star in the M35 field (Visual Mag 12.5). The finding won them first place at an international science competition (ISEF) in Pittsburgh. Meanwhile, the Kepler team heard the news about them and invited them to give a talk at NASA Ames. Scientists always love citizen science projects! They decided to help them confirm the exoplanet.

Exoplanets in open clusters are rare, especially Jupiter-sized giants. Why that is so is still a mystery. Exoplanets around A-type (hot) stars are even rarer. So NASA team helped them do followup observations using a 3.5-meter WIYN telescope on Kitt Peak (AZ) and later Keck (10m telescope in Hawaii)!. They did confirm the existence of the exoplanet. But to qualify that it is an M35 member, they had to prove that the star is indeed a member of M35 and not a foreground or a background star. The evidence, based on existing data was somewhat conclusive that it was indeed a member of M35. They submitted the paper, but the referee suggested waiting for definitive new data from the Gaia satellite to publish.

Gaia's precision data suggested that the star is not a member of M35 . This set them back and also they had to push back their publication. It is still an important discovery as the planet is around a hot star. The fact that it is not in M35 is also interesting as it means scientists have yet failed to find a giant planet in an open cluster.

It took 4 years from a citizen science project to a journal publication. The journey for them was worth it. Now they show off their planet/star at Astro-nights at UC Berkeley as they on the path of taking the hobby into a profession.

So there you have it--when you take your scope or binoculars out and see M35, there is a star lurking in your FOV which has now a known planet that has an SJAA connection!



Above: Shashank and Shishir in the control room of WIYN telescope.

Above: Shashank and Shihir near Kitt Peak Observatory, Arizona

New Loaner Items

Credit: Muditha Kanchana

SJAA Loaner Program recently added two refractors to it's collection.
Both of the below refractors were part of a single donation to SJAA. They are available for all SJAA members to try out.
https://www.sjaa.net/programs/loaner-telescope-program

Orion ED80T Refractor

This is a nice triplet refractor from Orion. It is lightweight and compact. Suitable as a grab and go scope. Can be an excellent imaging telescope (no camera accessories provided at this time)

- 80mm F/6 Carbon Fiber Triplet APO Refractor
- 480mm Focal length
- FPL-53 ED glass Air-spaced Triplet
- Dual speed Crayford focuser
- Carry Case with few other accessories.

Orion ED66 Refractor

This is a doublet ED refractor in the ultra lightweight telescope category. It can be used for night sky viewing as well as for terrestrial viewing thanks to the correct image diagonal included in the package.

- 66mm F/6 Carbon Fiber Refractor
- 400mm Focal length
- FPL-51 ED glass Air-spaced Doublet
- Dual speed Crayford focuser
- Carry Case with few other accessories.

Critical Deployment of NASA Webb's Secondary Mirror a Success

Credit: Nasa

In order to do groundbreaking science, NASA's James Webb Space Telescope must first perform an extremely choreographed series of deployments, extensions, and movements that bring the observatory to life shortly after launch. Too big to fit in any rocket available in its fully deployed form, Webb was engineered to intricately fold in on itself to achieve a much smaller size during transport.

Technicians and engineers recently tested a key part of this choreography by successfully commanding Webb to deploy the support structure that holds its secondary mirror in place. This is a critical milestone in preparing the observatory for its journey to orbit. The next time this will occur will be when Webb is in space, and on its way to gaze into the cosmos from a million miles away.

The secondary mirror is one of the most important pieces of equipment on the telescope, and is essential to the success of the mission. When deployed, this mirror will sit out in front of Webb's hexagonal primary mirrors, which form an iconic honeycomb-like shape. This smaller circular mirror serves an important role in collecting light from Webb's 18 primary mirrors into a focused beam. That beam is then sent down into the tertiary and fine steering mirrors, and finally to Webb's four powerful scientific instruments.

"The proper deployment and positioning of its secondary mirror is what makes this a telescope – without it, Webb would not be able to perform the revolutionary science we expect it to achieve. This successful deployment test is another significant step towards completing the final observatory," said Lee Feinberg, optical telescope element manager for Webb at NASA's Goddard Space Flight Center in Greenbelt, Maryland.

Though there are many preparations still underway for the full assembly of the James Webb Space Telescope's two halves, the secondary mirror test represents the last large milestone before the integration of Webb into its final form as a complete observatory. This operation was also another demonstration that the electronic connection between the spacecraft and the telescope is working properly, and is capable of delivering commands throughout the observatory as designed.

Webb will be the world's premier space science observatory. It will solve mysteries in our solar system, look beyond to distant worlds around other stars, and probe the mysterious structures and origins of our universe and our place in it. Webb is an international project led by NASA with its partners, ESA (European Space Agency) and the Canadian Space Agency.

As one NASA Webb's most important components, technicians and engineers thoroughly inspect the support structure that holds its secondary mirror in place following successful testing.

Credits: NASA/Chris Gunn

The Two-by-Three Chair

Credit: Muditha Kanchana

Most astronomy hobbyists have owned or seen a Vestil or a StarBound observing Chair. Few even use less common options like the Catsperch chair or DIY solutions like the Denver chair variants. Although above chairs are excellent performers, I have had my mind set on designing and building something different to meet my needs.

My design had several somewhat conflicting goals. Cheap to build yet doesn't look cheap. Lightweight but sturdy and comfortable. Easy to collapse and setup for transportation and field use. Safe and stable while allowing a full range of seat height adjustments.

I decided to use Kiln dried 2x3 whitewood studs that I was already very familiar with as the main structural material. They are $2 each for an 8ft long stud. A single 2x3 stud is sufficient to build the main body and the lateral supports. I used left over plywood to build the seat and the leg rest. I used two wood dowels on front and back sides of the 2x3 body to brace the seat. Used sandpaper backing ,in matching red color, on the back of 2x3 body as well as a dowel tightening knob to secure the seat from any accidental slides. The adjustable foot rest was done similar to the seat design using couple dowels. Some leftover pegboard scrap came in handy as a vented sliding backrest. I was planning to build a seat cushion. But, I found a memory foam seat that my wife didn't like which fitted around the 2x3 body quite well. An aluminum strip as a brace, few knobs from amazon, and nuts and bolts from home depot completed the rest of the components. Vertical 2x3 sections were attached to lateral support 2x3 sections with glue and hidden screws for extra stability. Two heavy duty ⅜" screws were used to attach the main 2x3 body piece to the back 2x3 support. These screws can be removed using a hex key to collapse the chair for transport. Under the seat there is a small compartment with a magnetic door to store the transport straps and hex key. A handhold slot was cut at the top top of the 2x3 body to easily move the chair around during an observing session. Decided to use fire-truck red and black colors to finish it all.

Chair was field tested few times. It easily supported ~240lb person. It felt stable both at the top and bottom seat positions and in-between. Overall, I was quite pleased with how closely it reached its original design goals.

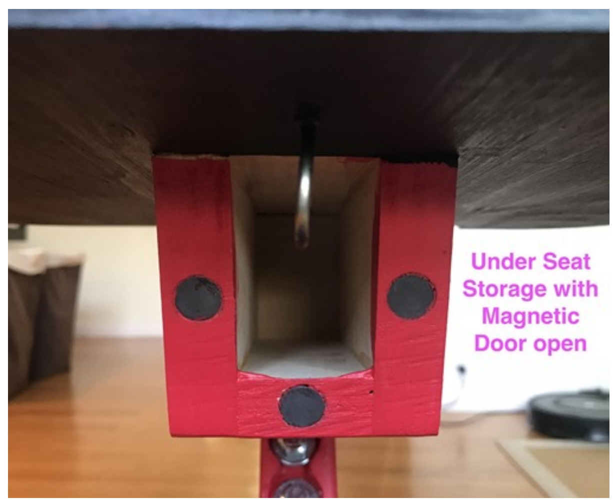
However, not all went well with this design and build. I accidentally pried one the wood inserts off the soft 2x3 wood while trying to dismantle the chair in a hurry. Repairing the damage was not easy. After this incident, I have been generally keeping the chair in the assembled position as most of the uses are in my backyard or at a nearby star party.

Any new build of a similar chair with softwood need to use a different method to secure wood inserts for frequent assembly-disassembly. This design take longer than planned to collapse and re-assemble. This is due to needing gentler assembly with softwood inserts and having to use a large hex key. Solid paint tend to make the sides of the seat and footrest stick to the 2x3 body when the chair stay unused for a while. Should have used colored stain or not painted the inner sides of the chair and footrest.

I am already thinking of future incarnations and improvements of this chair both as a simpler version and a taller stronger version. More likely to try a simpler version in the near future with weight reduction and ease of transport in mind.

Feature Summary:

- Seat height adjustable from 8” (bottom) to 32” (top)
- Independently adjustable backrest
- Adjustable footrest
- Lightweight (12.2lb without cushion)
- Collapsible
- High weight capacity (Tested for 240lb may be capable of more)
- Narrower front profile for ease of use next to tripods and scope bases (lateral balance mostly from the back horizontal brace)



SJAA Member Astrophoto Gallery

Date: July 6, 2019

Location: Pinnacles East, Paicines, CA

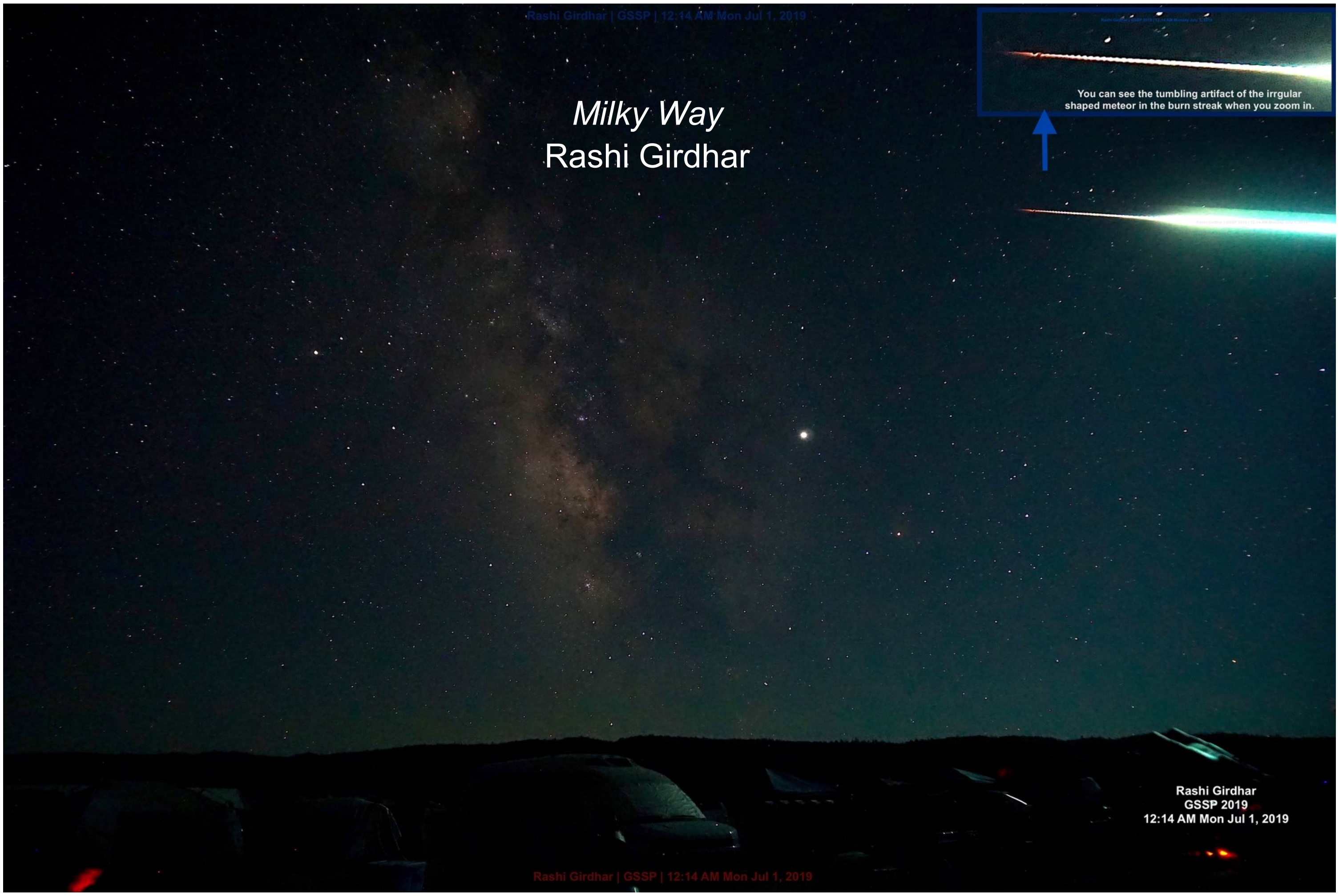
Imaging Telescope: Astro-Tech AT65EDQ

Imaging Camera: Nikon D5500 (Ha Mod)

Mount: iOptron iEQ45 Pro

Software: Pleiades Astrophoto PixInsight 1.8 Ripely

SJAA Member Astrophoto Gallery



Dates: July 1, 2019

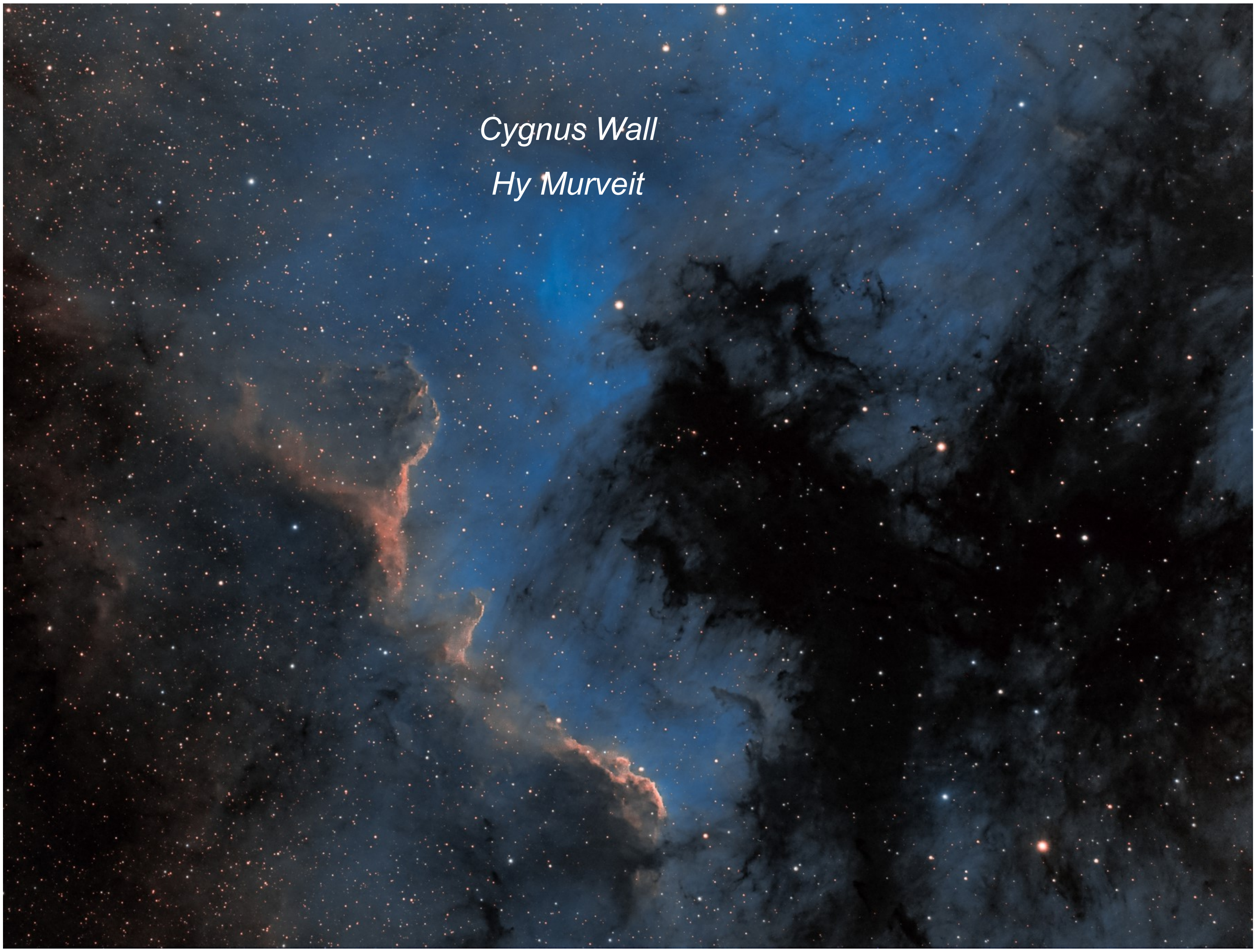
Location: Golden State Star Party, Adin

Telescope: N/A

Camera: Sony A6000 with stock 16mm-50mm EOS lens @ 16mm Focal Length

Mount: Orion Paragon Tripod with pan head (non-tracking)

Software: On Camera HDR function + Basic Lighting and Contrast adjustment



Date: June 29, 2019

Location: Golden State Star Party

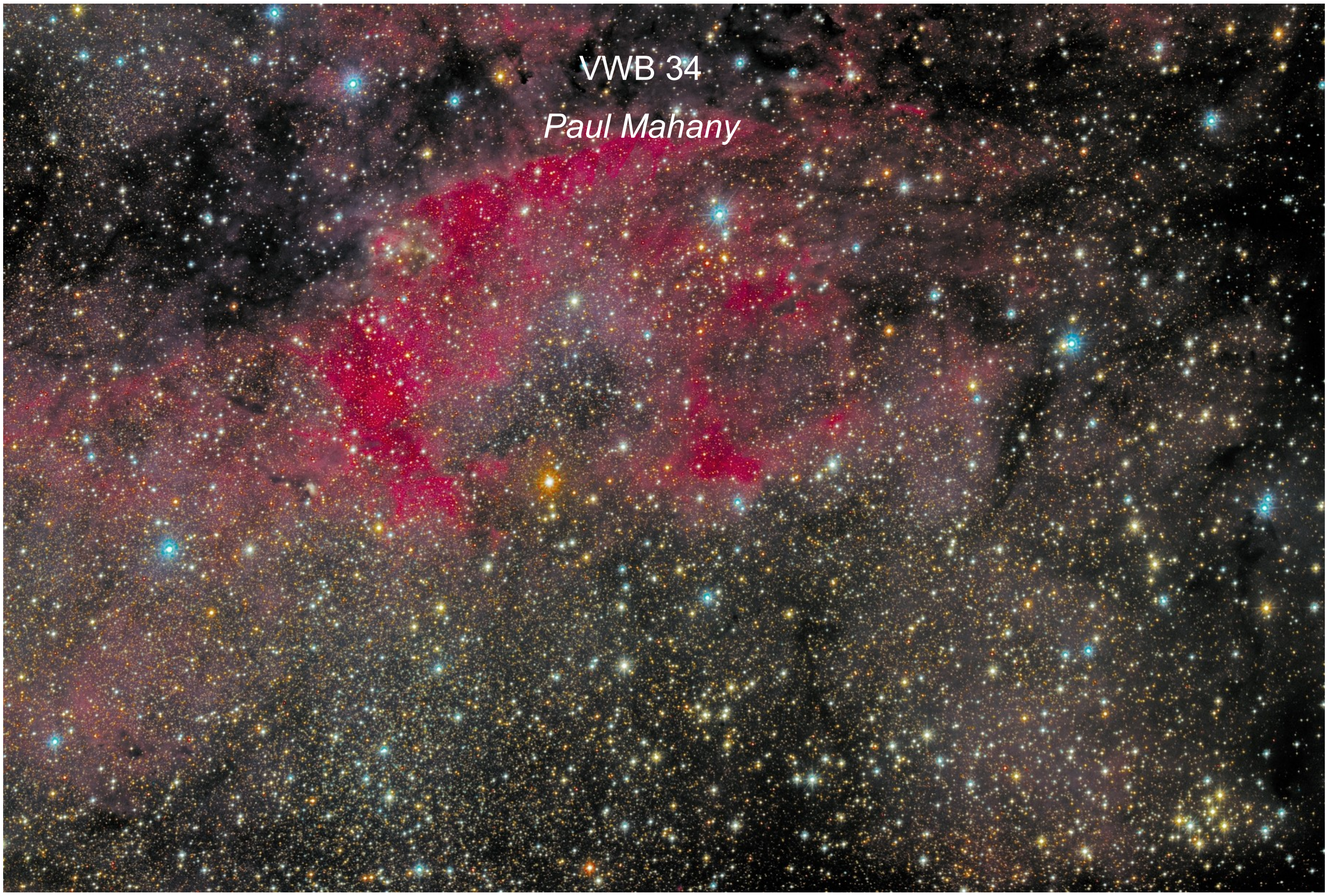
Telescope: Williams Optics Zenithstar 105 ED Triplet APO

Camera: ZWO ASI 600mm Pro Cooled

Mount: Orion Atlas AZ/EQ Pro

Software: Adobe Lightroom CC Classic, Adobe Photoshop CC, KStars Ekos/INDI, Stellarmate OS

SJAA Member Astrophoto Gallery



Date: July 6 2019

Location: Manzanita Observatory, near Yosemite National Park

Telescope: Celestron RASA 11”

Camera: Orion StarShoot Autoguider

Mount: Software Bisque Paramount MX+

Software: TheSkyX Pro, PixInsight1.8, SkyTools, BYEOS, PHD Guiding 2 PHD 2 Guiding, StarTools

Kid Spot

Kid Spot Jokes:

- **What does a cow like to look at in a sky?**
  The mooooon
- **An Alien said ants must be big**
  The human responded Ant-are-s mall
- **What kind of monitor do you use to attach to your telescope?**
  Ve Ga

School Star Party Scheduling Has Begun

Credit: Cat Cvengros

The SJAA School Star Party scheduling has begun. There are two action items for our club: 1) Reach out to Cat to let her know if you would like to join the esteemed School Star Party volunteer astronomers! It is THE BEST THING to hear a kid or parent exclaim in wonder the first time they see a crater of the moon, the rings of Saturn or the amazement of the double star Albireo.

2) Forward this link to teachers or administrators to get on the books: https://www.sjaa.net/programs/school-star-party/want-a-school-party/ or simply Google "school star party San Jose." The website has many tips for first time star party hosts, and also reminders for the seasoned partiers. Due to the high number of cancellations from the rain this spring, we are prioritizing the schools that did not get scheduled. Thank you for your interest and patience!

Yosemite Star Party

Credit: Ken Miura

This year's Yosemite Star Party took place on the evenings of July 26 and 27.

I coordinated the event for the first time this year. We lucked out in selecting the dates: the moon was past the last quarter and did not come up until 1 am or so. We had 10 participants this year, but two could not make it.

We had been notified by the park ranger's office that the Bridalveil campground was still under maintenance and not available. At first, we were offered the Wawona campground like last year, but then it was changed to the Yellow Pines campground in the Yosemite Valley, which is meant for volunteers and very small in size (approximately only 8 or 9 tents maximum.) It also took more than one hour to drive up to the Glacier Point from there. The next morning, we were offered the choice of the Bridalveil campground which just opened. Because of the inconveniently long drive on the previous night, we chose to move to Bridalveil.

The sky was very dark without the moon. We had about 10 telescopes and binoculars and had an excellent time observing the planets, galaxies, double star, globular clusters etc. In particular, the Saturn just passed the opposition 10 days ago. But there was the issue of a bright light outside the Gift Shop in the north direction, which never existed in the past. So, on the second evening I complained to the park ranger in charge, and he put a temporary cover over it. It was much better than the previous night but still not completely dark. This may continue to be an issue in the future. I wonder how other clubs reacted to it.

Above: Group Photo

Credit: Ken Miura

From the Board of Directors

Announcements

- ◊ QuickSTARt program is currently on hold, pending some restructuring
- ◊ SJAA Library is operational and available to all members during most Houge Park events
- ◊ We need more volunteers to step forward and help with various club programs and events. Please contact Swami at president@sjaa.net if you are interested to help the club out and for additional information regarding this.

Board Meeting Excerpts

June 15, 2019

In attendance
Swami, Sukhada, Glenn, Ken, Vini, Sandy, Wolf
Excused Absences: Rob J, Rob C
Guests: Marianne, Kaushal, Jai, Kanch

Houge Park Facility Cleanup and storage reshuffle
Kanch discussed his proposal for reorganization and cleanup with the board.

Preschool lights
Vini met with the Cambrian district facilities team and they were going to provide us with physical access to the grounds and allow us to cover the light fixtures for ITSP events

SJAA Collaborating with USS Hornet Museum
SJAA is collaborating with USS Hornet Museum to join in their celebration of 50th anniversary of Apollo 11 mission, Splashdown 50

Outreach Requests
A few outreach requests have been received by SJAA from Kelley Park, and San Jose Park Rangers and Montalvo Arts Center.

July 20, 2019

In attendance
Swami, Sandy, Vini, Glenn, Sukhada, Wolf , Vini, Rob J, Ken
Excused Absences: Rob C
Guests: Peter, Milind, Kaushal

Outreach Request Policy
Sandy is working on an official SJAA policy for outreach requests received by SJAA from various sources.

USS Hornet museum
SJAA participated in Splashdown 50, the 50th anniversary celebration of the Apollo 11 mission organized by the USS Hornet Museum. Ken Miura setup a night telescope on the flight deck on July 19, and the Solar Astronomy team setup solar telescopes on July 20. Both event were very popular, and there was a steady stream of visitors enjoying the views through our telescopes.

Stars Over Yosemite
Stars Over Yosemite star party is scheduled for July 26-27. Ken Miura is coordinating the event.

Preschool lights during ITSP
Cambrian School district has allowed us to cover up the lights during the ITSP sessions. That had made the Houge Park parking lot much darker and improved the viewing conditions during ITSP. However, at least one of the shades has gone missing. They need to be collected as soon as the star party is over. Vini will be training others on how to remove and install the shades.

Pending Outreach Requests
Starry Nights at Montalvo Arts Center on August 9, Perseids Meteor Shower on August 10 with San Jose Park Rangers, City of Cupertino Astronomy Night on August 17, Cambrian Festival on August 25.

August 17, 2019

In attendance
Swami, Wolf, Vini, Rob J, Glenn, Sandy, Ken
Excused Absences: Rob C, Sukhada
Guests: Marianne , Peter M

Outreach Policy
The board discussed the Outreach Policy and made a few minor revisions. The final policy will soon be published to the public.

2020 Events Calendar planning
Planning for the Events calendar for 2020 has started. Houge Park will be unavailable for 11 days in February and November 2020 to accommodate the Registrar of Voters for the California primary elections and the Presidential elections.

Internship program
SJAA is brainstorming ideas for instituting an internship program for astronomy student enthusiasts. Details to be worked out in the coming few months.

Recruiting volunteers
Need to recruit more volunteers to help out with various programs and other SJAA tasks. Swami will be requesting all current leaders to write a short job description of their roles at SJAA, which can be used to educate other members of the required duties.

Past Events
Events at Montalvo Arts Center, Perseids at Alum Rock Park, and Stars Over Yosemite were all very successful, and drew lots of oohs and aahs from the crowds. For the Stars Over Yosemite event summary and group photo please look at Page 12.

SJAA Library

SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

http://www.sjaa.net/programs/fixit/

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

http://www.meetup.com/SJ-Astronomy/

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick STARt program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

http://www.sjaa.net/programs/quick-start/

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

http://www.sjaa.net/programs/beginners-astronomy/

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.
If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.
http://www.sjaa.net/programs/loaner-telescope-program/

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

http://www.sjaa.net/programs/imaging-sig/

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.
Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

http://www.sjaa.net/programs/school-star-party/

SJAA Contacts

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Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Sandy Mohan
Director:	Rob Chapman
Director:	Vini Carter
Director:	Ken Miura
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
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Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Quick STARt	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Media Manager:	Jessica Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki

E-mails: http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
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San Jose, CA 95159-8243
http://www.sjaa.net/contact

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here

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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **$20** Regular Membership with **online** Ephemeris

☐ **$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: **http://www.sjaa.net/sjaa-newsletter-ephemeris/**

Questions? Send e-mail to: **memberships@sjaa.net**

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: **http://www.sjaa.net/join-the-sjaa/**

Name: ______________________________

Address: ______________________________

City/ST/Zip: ______________________________

Phone: ______________________________

E-mail address: ______________________________